AI ACADEMY · LEVEL 2 · GRADE 7 · WEEK 15 · STUDENT LESSON

Choose Informative Questions

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

Before you start

Week 14: a useful question separates target groups.

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Fit a root question and leaf predictions from training examples. Try the recall before opening notes.

Your learning target

I can fit a **root** question and **leaf prediction**s from **training** examples.

I can trace a new example through the learned tree.

I can compare a shallow tree with a baseline and state its limits.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Decision Trees. Give an example and a mistake that the idea helps you avoid.

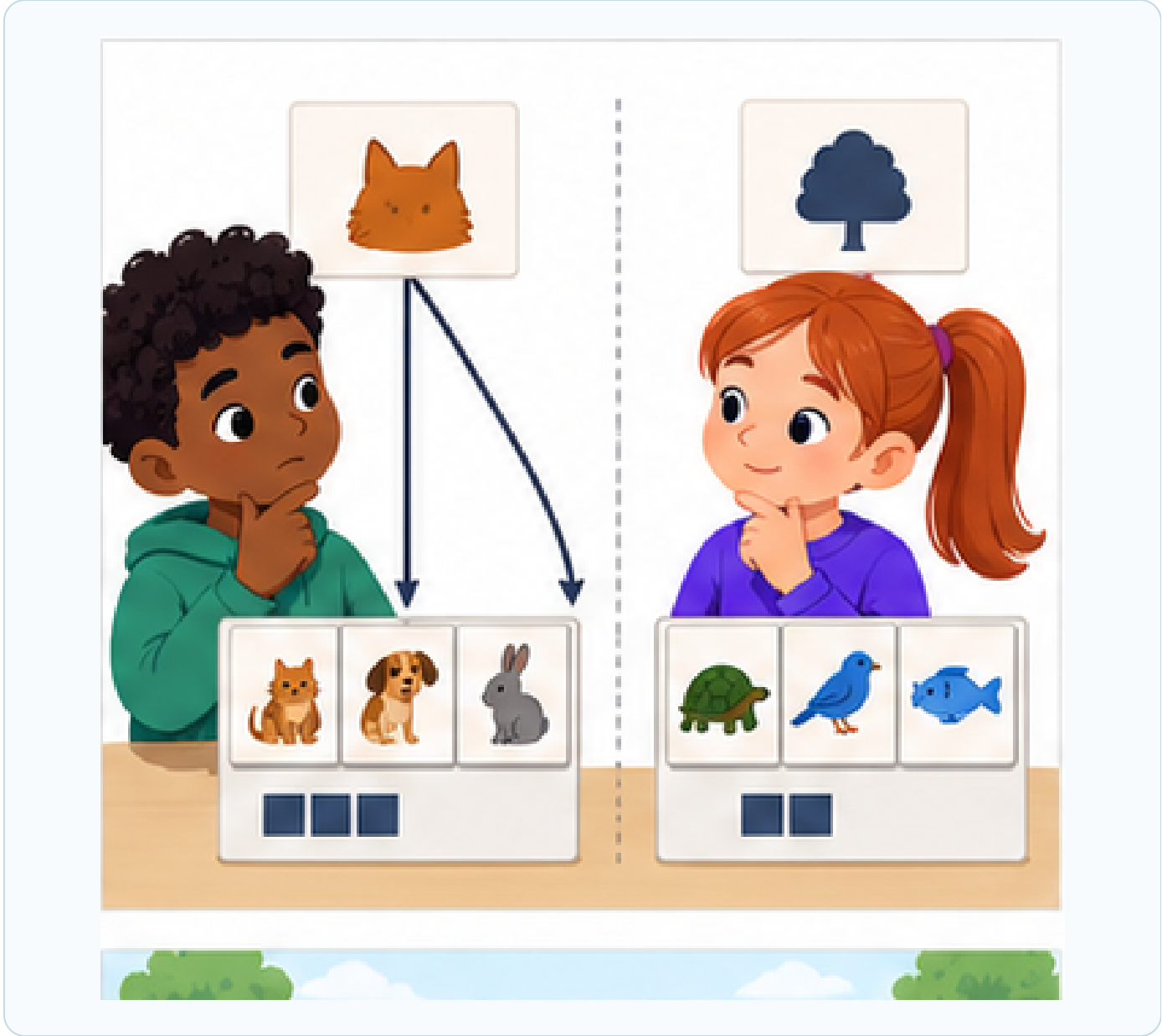
Without opening the old lesson, explain one key idea from Accuracy as a Fraction. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Features and Distractors. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Week 15: Choose **Informative** Questions

A strong question eliminates many possibilities without relying on accidents.



THINKER MISSION

Compare two first questions by **remaining possibilities**.

Thinker name and date:

> **Write or draw your reasoning and evidence.**

Readiness for the connected explanation

> Why does a cutoff with zero **training** errors still need validation on separate examples?

This week's end goal

By the end, I can choose a question that separates possibilities rather than repeating a constant clue.

STUDENT LESSON · 2 OF 18

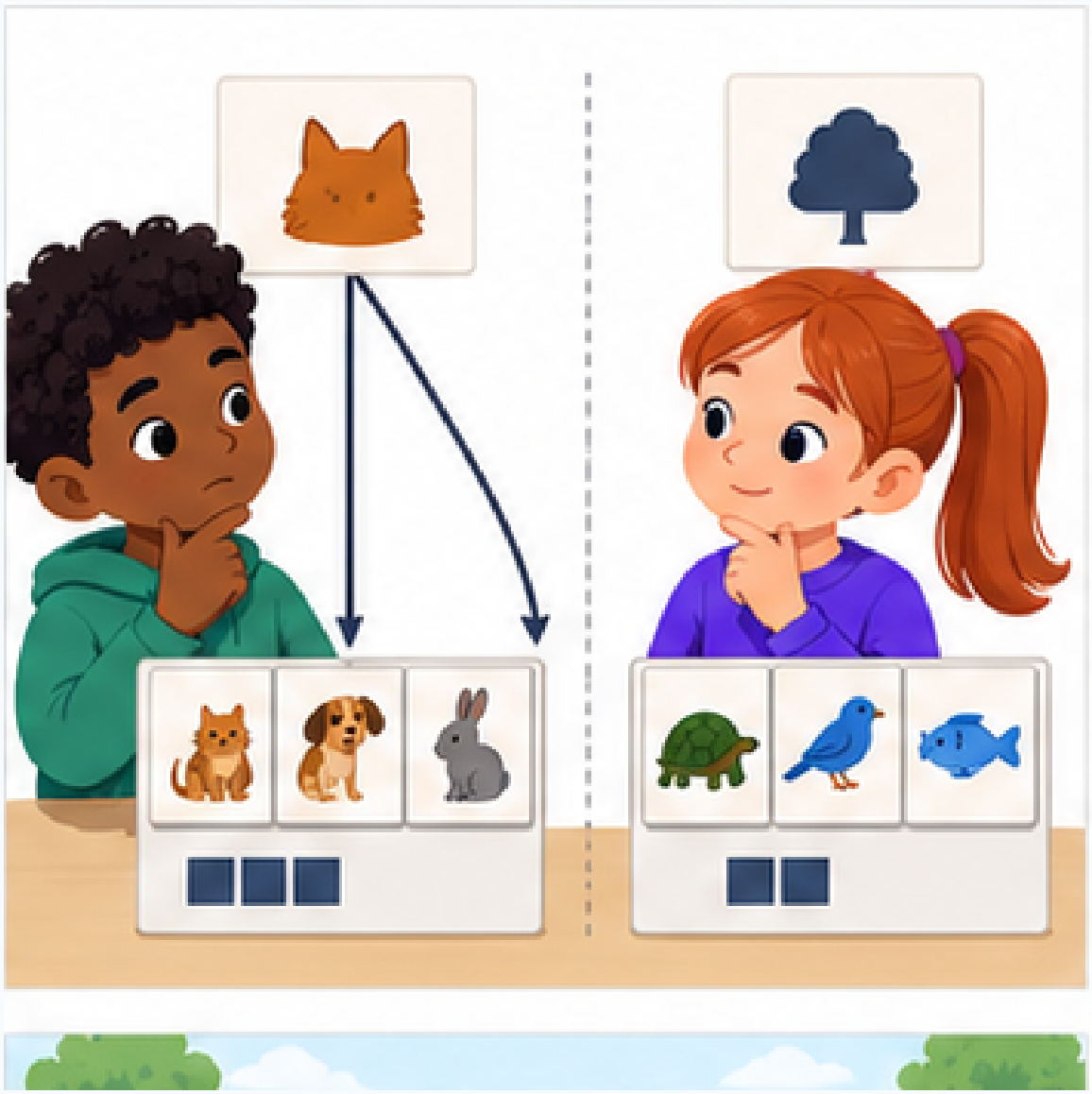
Notice the evidence

> Investigate the supplied Choose Informative Questions case. Separate observations, given facts and assumptions before choosing an explanation.

PICTURE STORY

Notice → Predict → Explain

Begin with visible **evidence**. Do not explain more than the picture and information support.



Start with this puzzle

MYSTERY

The learners must use choose **informative** questions to solve a problem. What information do they have? What decision or calculation do they make? What would count as **evidence** that it worked?

Three **observation**s (what you can point to):

> **Write or draw your reasoning and evidence.**

My **prediction** and the clue supporting it:

> **Write or draw your reasoning and evidence.**

A closer explanation

A decision tree asks a feature question and follows a **branch** to a **prediction**. A tree with one question has **depth** one and is sometimes called a stump.

Unlike Week 8, where the two **output** labels were fixed in advance, this week we fit both the question and the label stored at each **leaf**.

Our six fictional **training** rows have two binary sensor flags available before **prediction**: near and tall. A value of 1 means that flag is present; 0 means absent. The reviewed scene label is separate.

Each row represents a different trip on a familiar route, including rows with identical values. Similar readings can have different scene labels. These are teaching examples, not robot control instructions.

Row	Near	Tall	Label
1	1	0	blocked
2	1	1	blocked
3	1	0	blocked

Row	Near	Tall	Label
4	0	1	clear
5	0	0	clear
6	0	1	blocked

Try near first and tall second. For each question, group the **training** rows by 0 and 1, then predict each group's majority label; choose clear for tied label counts.

Count all **training** mistakes and keep the first question if scores tie. Stop after one question. This is our stated classroom **fitting** procedure.

STUDENT LESSON · 3 OF 18

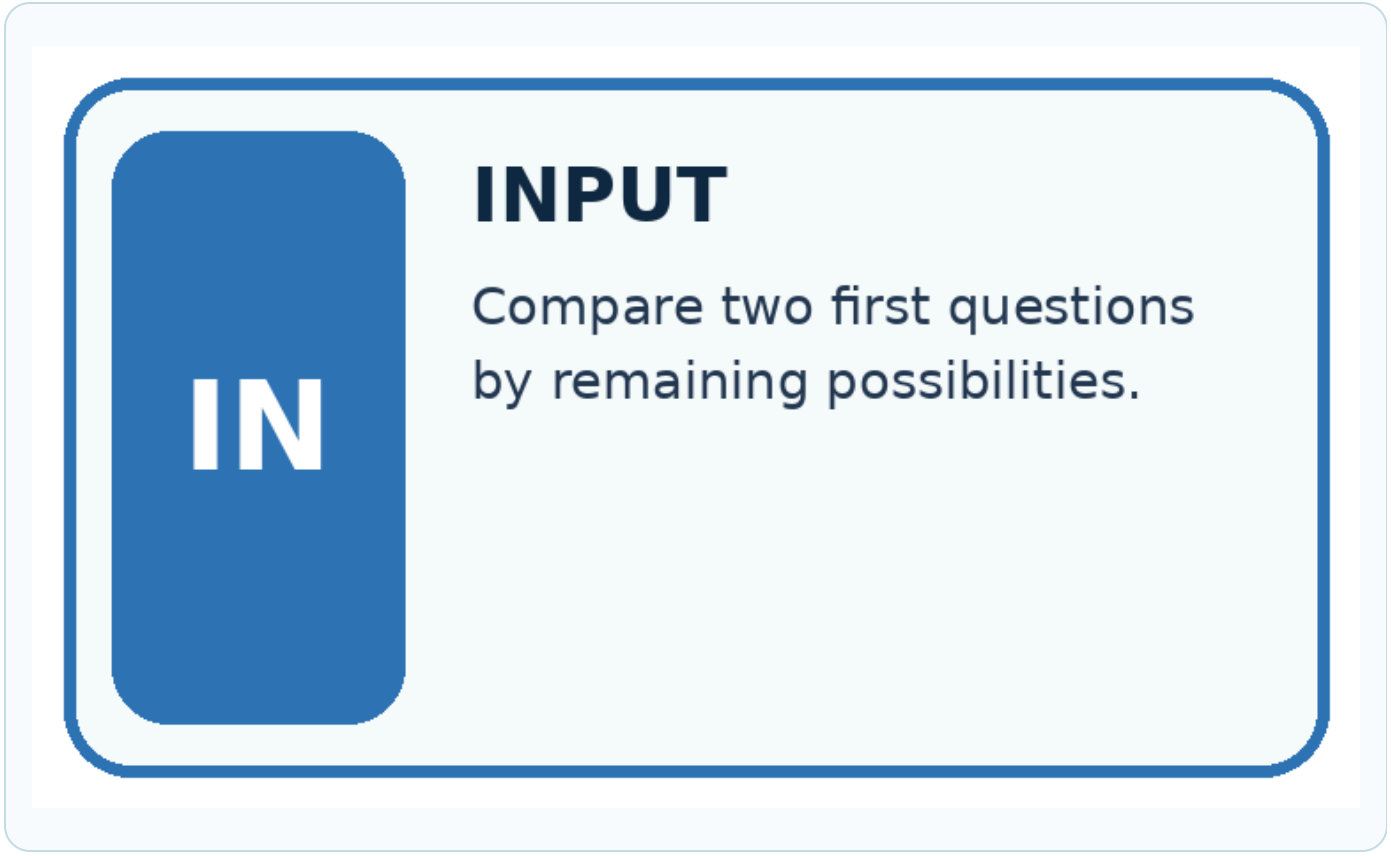
See how it works

> Follow the mechanism for Choose Informative Questions. Connect each labelled input, deciding step and result.

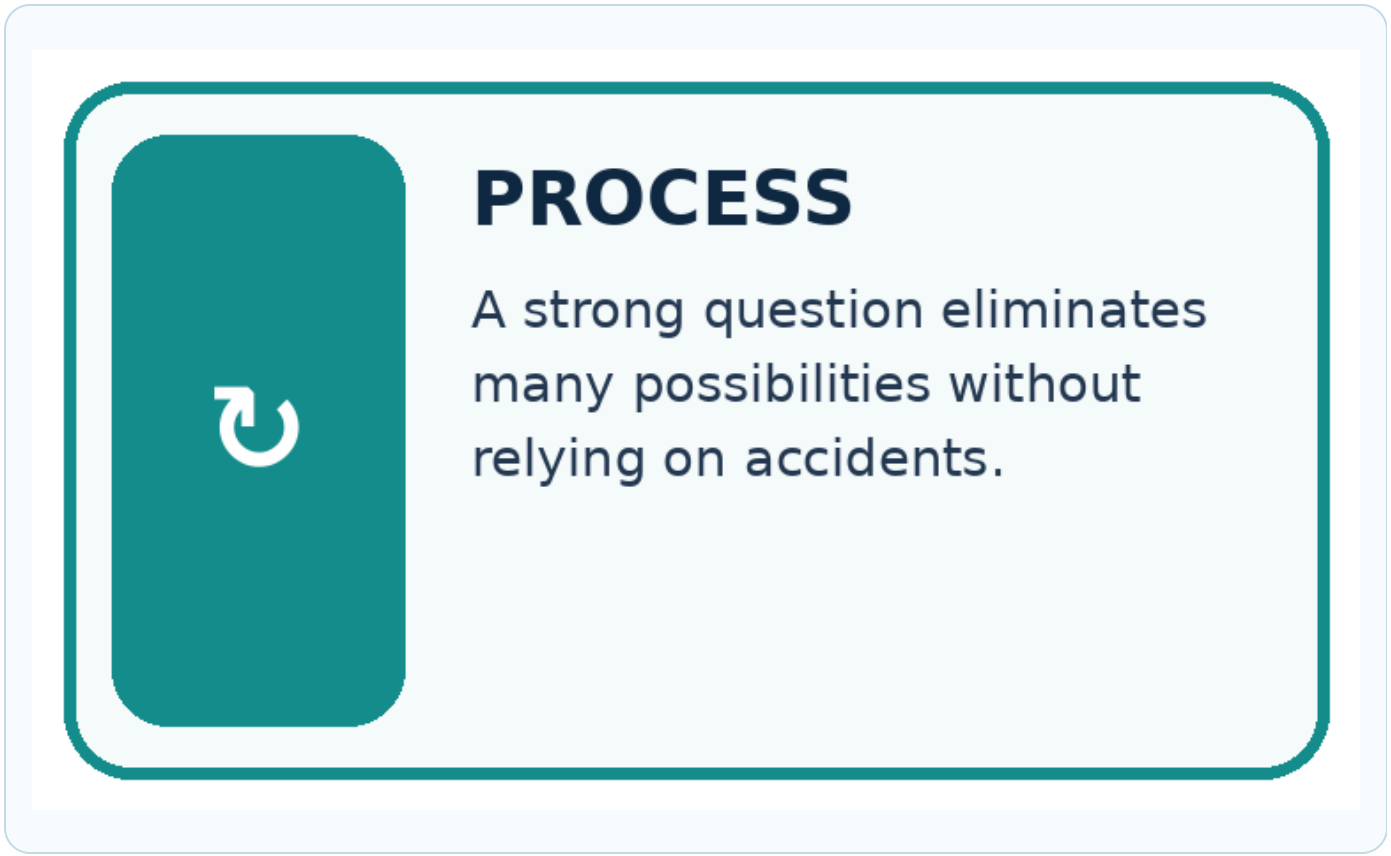
VISUAL MECHANISM

Input → **Process** → **Output**

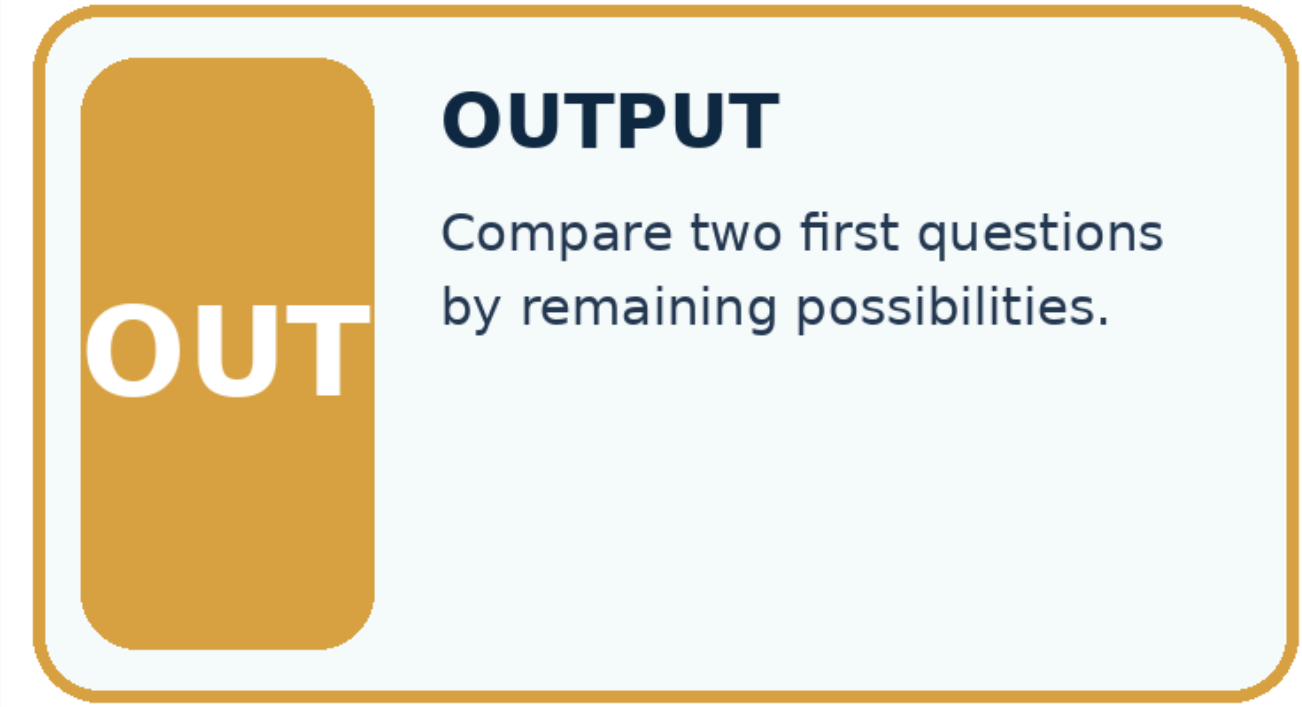
Trace the information. At each stage, ask what changed and how you could test it.


IN
INPUT
Compare two first questions by remaining possibilities.

↓


PROCESS
A strong question eliminates many possibilities without relying on accidents.

↓



Which stage could fail even when the other two are correct? Explain:

Write or draw your reasoning and evidence.

A closer explanation

```
rows = [(1, 0, "blocked"), (1, 1, "blocked"),
        (1, 0, "blocked"), (0, 1, "clear"),
        (0, 0, "clear"), (0, 1, "blocked")]
def majority(labels):
    if labels.count("blocked") > labels.count("clear"):
        return "blocked"
    return "clear"
best_errors = len(rows) + 1
```

```
for feature in [0, 1]:
    leaves = {}
    for value in [0, 1]:
        labels = [row[2] for row in rows
                  if row[feature] == value]
        leaves[value] = majority(labels)
    errors = 0
    for row in rows:
        if leaves[row[feature]] != row[2]:
            errors = errors + 1
    print(feature, errors)
    if errors < best_errors:
        best_errors = errors
        tree = (feature, leaves)
print(tree[0], tree[1][0], tree[1][1])
```

Feature 0 is near; feature 1 is tall; row[2] is the label. **Output** is 0 1, then 1 2, then 0 clear blocked. Near makes one **training** mistake: near=0 contains two clear labels and one blocked, so that **leaf** predicts clear.

Near=1 contains three blocked labels. Tall makes two mistakes because both its groups have a blocked majority.

The stored tree contains the chosen feature and its two learned leaves. The code assumes nonempty data, binary flags, allowed labels and examples on both **branch**es of each candidate.

An empty **branch** would need a declared fallback; this small example does not implement one. The score is total **training** mistakes, not an average giving small and large **branch**es equal weight.

This is an explicitly limited **depth**-one learner. General tree algorithms can repeat **split** selection within **branch**es and often use impurity criteria such as Gini. Our code does not implement that full procedure.

Keeping the scope small lets you inspect every decision.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

Compare the cases

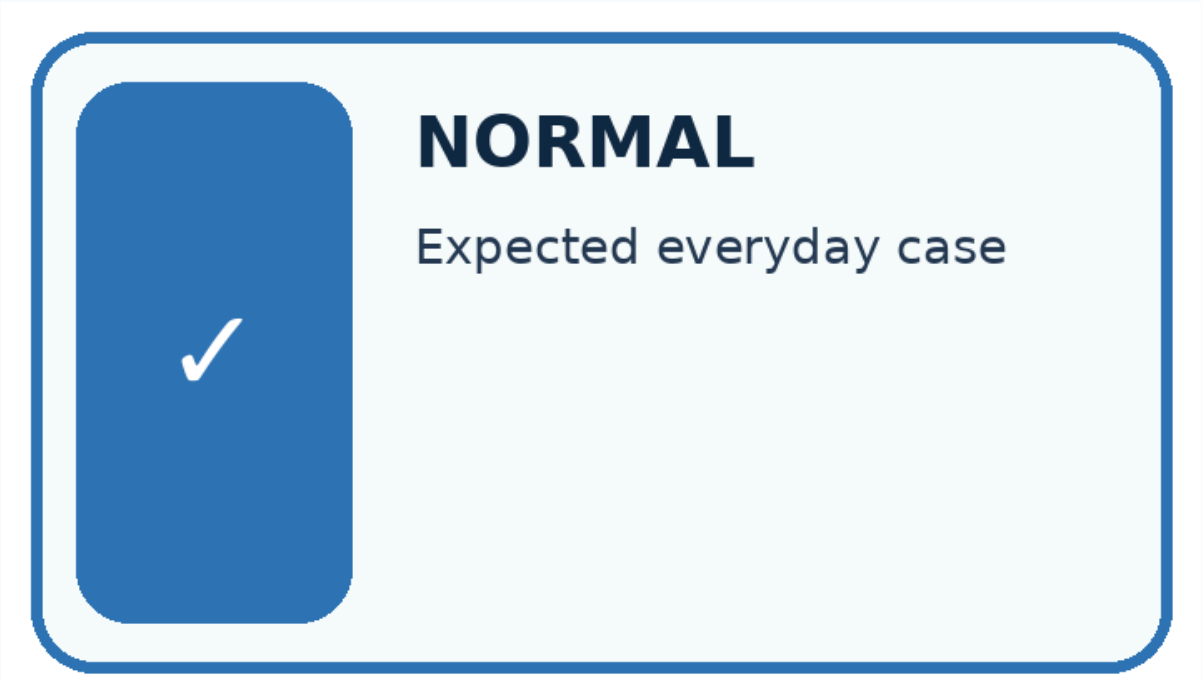
> Compare the supplied cases for Choose Informative Questions. Identify the detail that changes the conclusion.

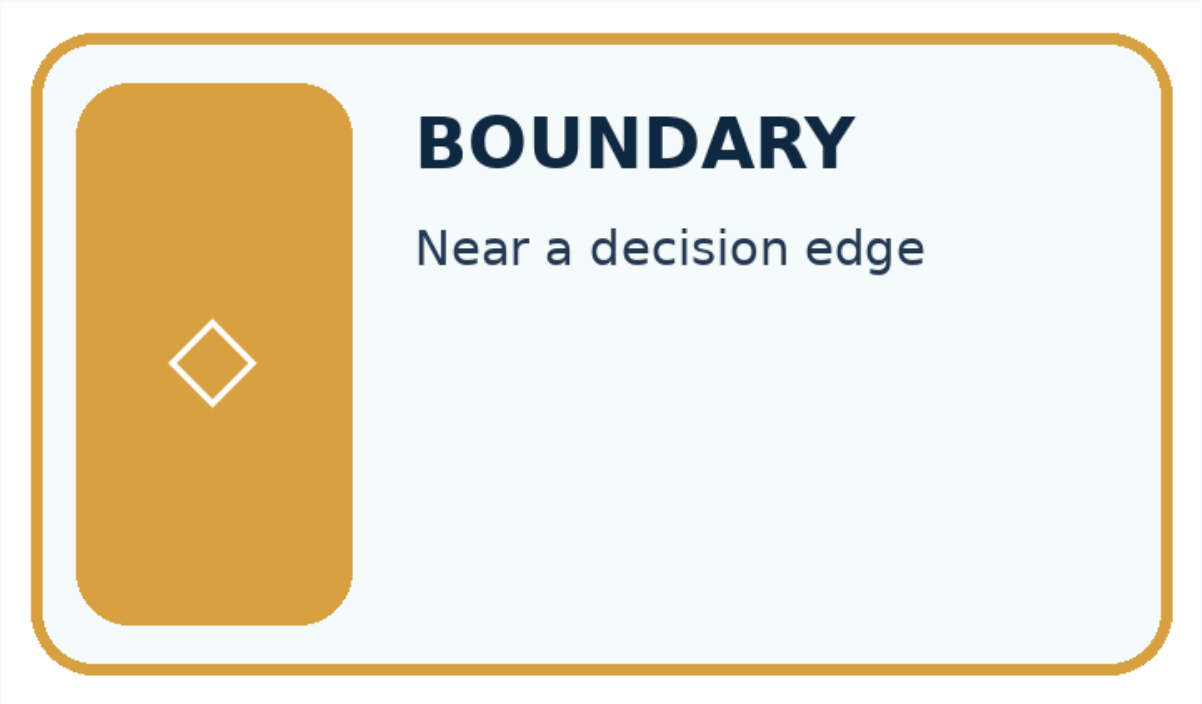
The case to investigate

CASE GALLERY

Four Tests, Four Kinds of **Evidence**

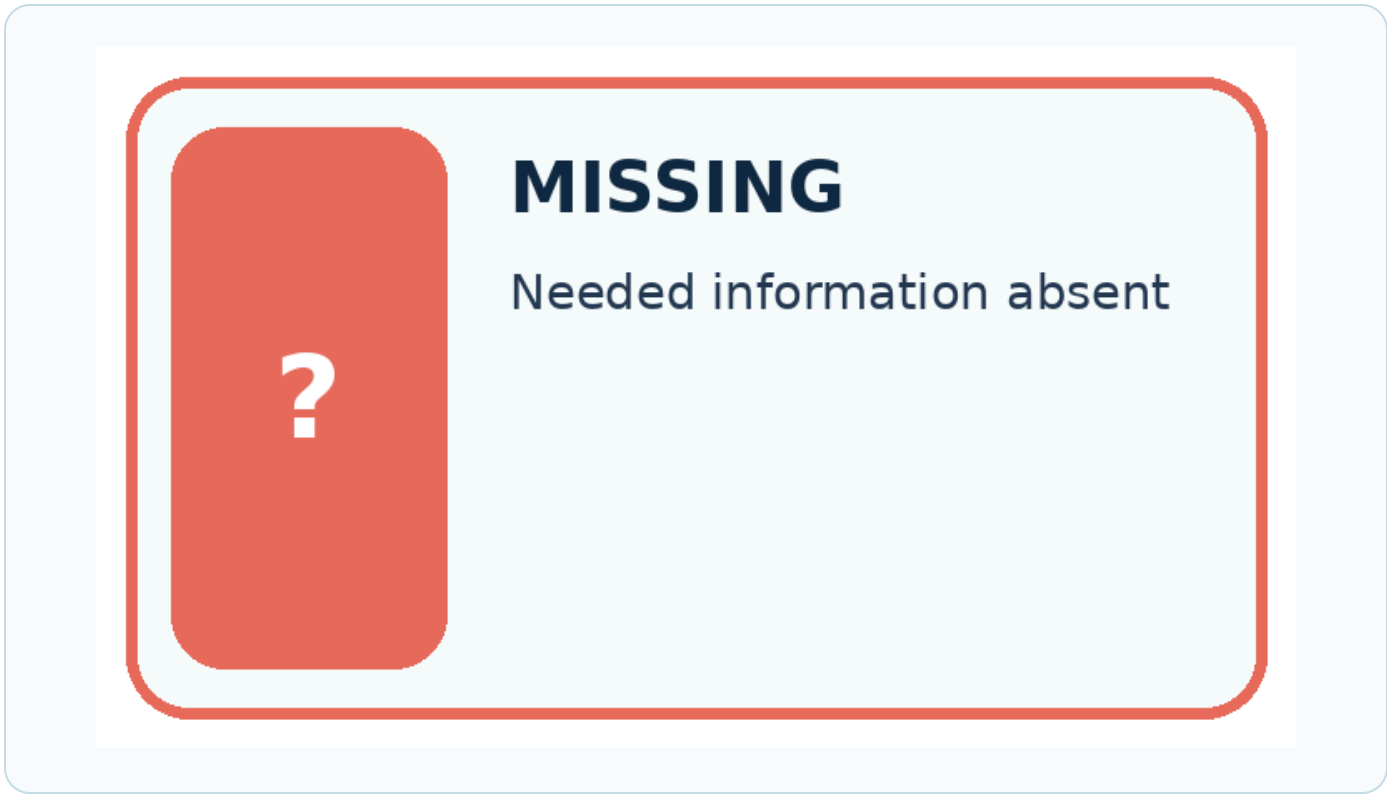
A system is not understood until it survives more than one comfortable example.





The normal case

NORMAL CASE: predict and name **evidence**. BOUNDARY CASE: predict and name **evidence**.


?
MISSING
Needed information absent


→
TRANSFER
Fresh situation or user

Check missing information

MISSING INFORMATION: predict and name **evidence**. TRANSFER CASE: predict and name **evidence**.

Which case is most likely to reveal a hidden assumption? Why?

Write or draw your reasoning and evidence.

Change one thing in the pictured case

Change one thing

Change: The current possibility set contains only the four mystery cards.

Predicted result

Prediction: Asking mystery now leaves all four; it no longer narrows the set.

Why the result follows

Reason: Question value depends on the remaining candidates, not just on its wording.

Words that unlock the idea

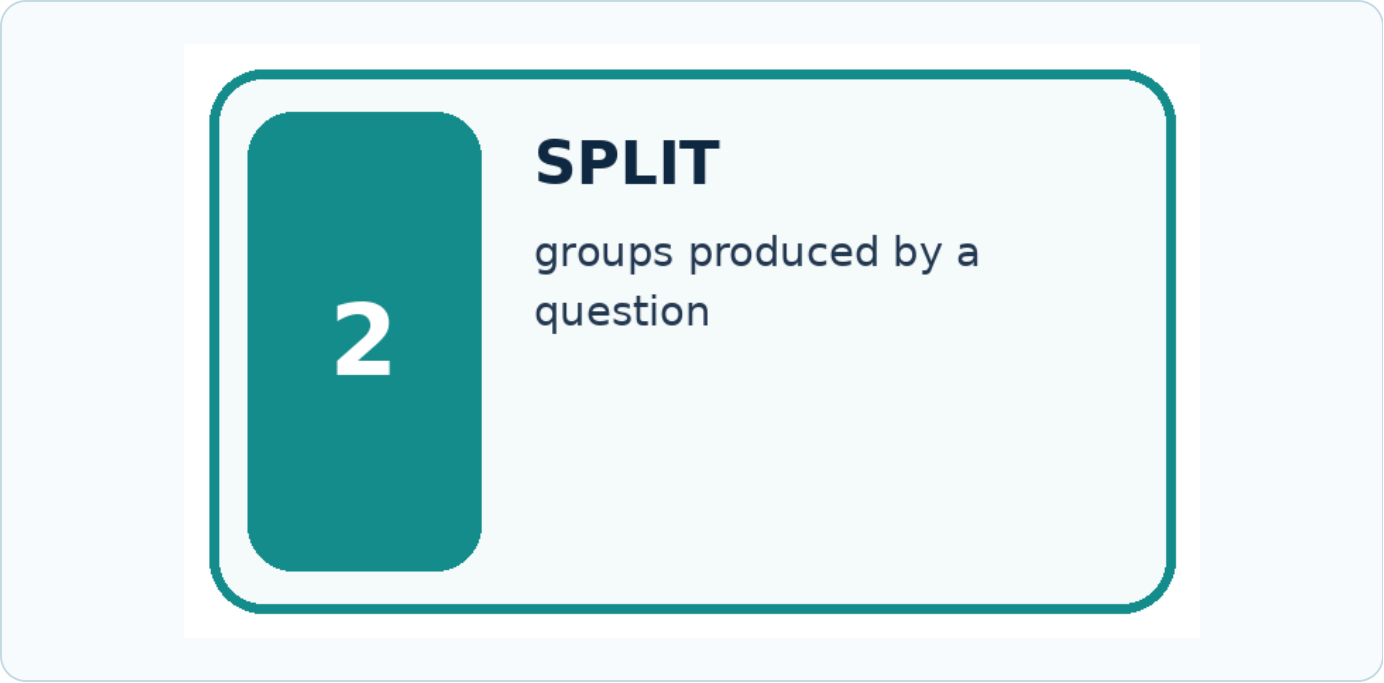
> Use the vocabulary of Choose Informative Questions to explain a concrete decision, rather than only memorising definitions.

VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.




2
SPLIT
groups produced by a question


3
REMAINING POSSIBI
candidates still possible



Word	My own example	Not this / boundary
informative	Providing evidence that helps distinguish relevant possibilities or reduce uncertainty.	
split	Separate a dataset into portions with different development or evaluation roles.	
remaining possibilities	The options still consistent with the evidence after some have been ruled out.	
efficiency	The resources required to achieve a result, such as time, memory or number of steps.	

Words used in the closer explanation

Word	Meaning
Root	The first question asked by a tree
Branch	A path selected by the answer to a question
Leaf	An endpoint that stores a prediction

Word	Meaning
Depth	The largest number of question steps from root to leaf

STUDENT LESSON · 6 OF 18

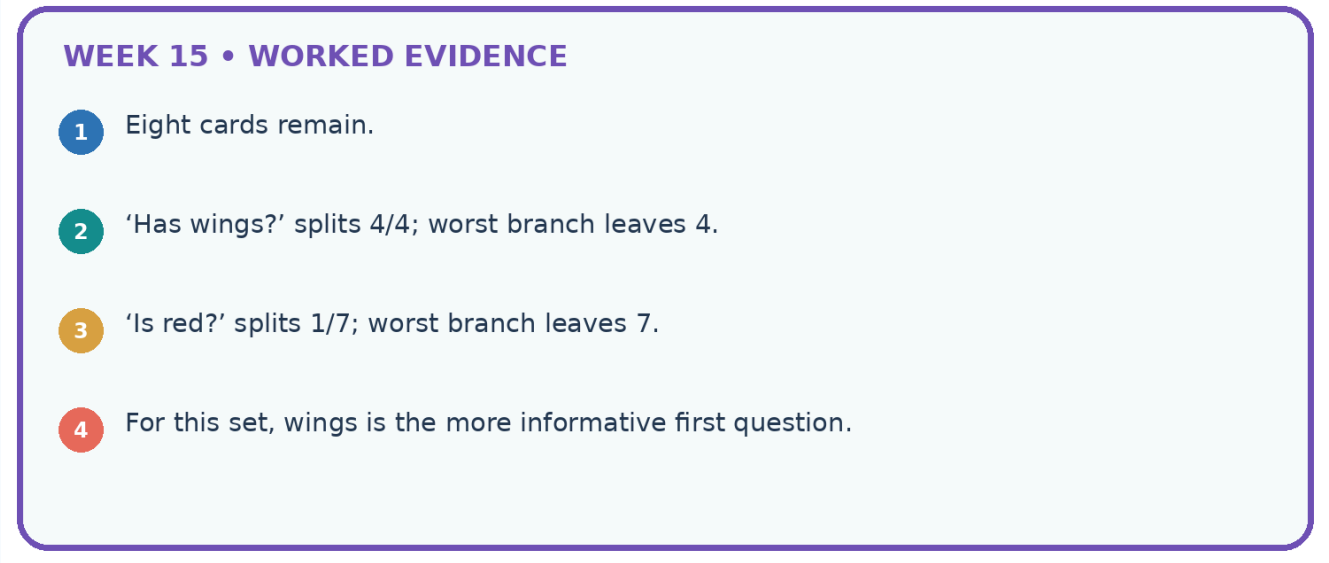
Follow a worked example

> Follow the worked trace for Choose Informative Questions, predict a step, then compare with its explanation.

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.



1. Eight cards remain.

2. ‘Has wings?’ **split**s 4/4; worst **branch** leaves 4.

3. ‘Is red?’ **split**s 1/7; worst **branch** leaves 7.

4. For this set, wings is the more **informative** first question.

Recalculate, retrace, or restate the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

Two paths through the learned tree

ROOT QUESTION	BRANCH ANSWER	LEARNED LEAF
Is near equal to 1	No near is 0	Predict clear
Is near equal to 1	Yes near is 1	Predict blocked

Both paths start at the same root. Tall is not used by this fitted tree; the leaf labels came from training counts.

Worked explanation · Choose Informative Questions

Validation rows are (near=1,tall=1,blocked), (0,0,clear), (0,1,blocked), (1,0,clear). Follow only near: **prediction**s are blocked, clear, clear, blocked. Two of four match.

The **training**-majority baseline predicts blocked, since **training** has four blocked and two clear; it also matches two of four validation labels.

Which validation rows does the tree miss, and which does the baseline miss? Explain why the tall value cannot change this tree's prediction.

Keep the fitted **root** and leaves unchanged while scoring. Equal accuracy does not mean identical errors, and five correct **training prediction**s out of six is not the validation result.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Choose Informative Questions

A fictional guessing game has eight catalogue cards and asks yes/no questions to narrow the target.

The facts we will use

Supplied facts

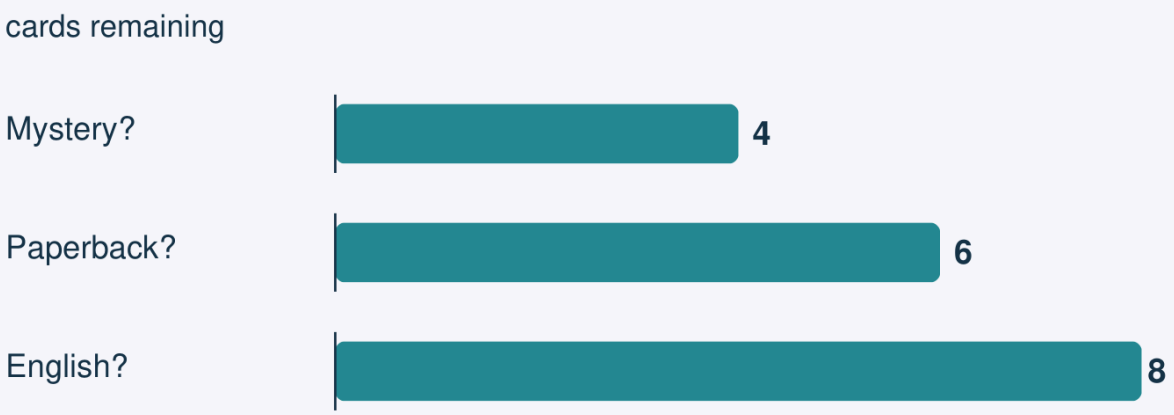
Cards: four mystery, four science; six paperback, two hardback; all English.

Supplied facts

Compare worst-case remaining group size after asking mystery?, paperback? or English?. No claim about unequal target probabilities is made.

AI ACADEMY / AI-2/15

Worst-case remaining candidates



Bar length shows the supplied value; the marked baseline is zero.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-2/15

Choose Informative Questions

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. List possibilities

There are eight possible target cards initially.



2. Mystery question

Yes group 4; No group 4; largest remaining group 4.



3. Paperback question

Yes 6; No 2; largest remaining group 6.



4. English question

Yes 8; No 0; largest remaining group 8.

CHANGE ONE THING

The current possibility set contains only the four mystery cards.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. List possibilities	There are eight possible target cards initially.
2. Mystery question	Yes group 4; No group 4; largest remaining group 4.
3. Paperback question	Yes 6; No 2; largest remaining group 6.
4. English question	Yes 8; No 0; largest remaining group 8.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Under the stated worst-case criterion, mystery is the most useful of these questions because it leaves at most four cards. The constant language clue never narrows the set.

This is an identification-game criterion; a classifier **fitting** labels may instead compare **prediction** errors or another declared **training** measure.

What this example does not establish

Keep the stated limit

Balanced groups do not guarantee a truthful user answer or optimal choices under every probability or cost model.

Keep the stated limit

Read the labelled picture: Choose the best original question and explain why its more balanced **split** matters here.

Keep the stated limit

Change one thing: The current possibility set contains only the four mystery cards. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Choose Informative Questions claim and explain a repair using the supplied evidence.

Find the first error

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak **evidence**.

!

COMPARE • DIAGNOSE

Weak idea: The most unusual question is always the most informative. Strong move: locate evidence, change one cause, and retest.

Evidence record	Expected	Actual	Likely cause / next test
normal case	works		
boundary case	stated rule		
missing input	safe response		
fresh transfer	generalizes		

Repair one cause. Why is this repair better than changing several things?

Write or draw your reasoning and evidence.

A closer explanation

Record the two candidate questions, **branch** label counts, total errors, tie rules, **depth** limit, stored tree and validation **prediction**s.

Retain all rows, including the blocked example in the near=0 **branch**; removing a disagreement would change the **evidence**.

Check the claim against the facts

Our tree uses a single feature because that question scored better under this procedure. This neither establishes that near causes blockage nor that tall is useless in every dataset.

Next week a nearest-neighbour method will predict by comparing examples instead of following learned questions.

A learner says the tree is reliable because near=1 has only blocked training labels. Rewrite this claim using the leaf evidence, validation score and one limitation.

STUDENT LESSON · 8 OF 18

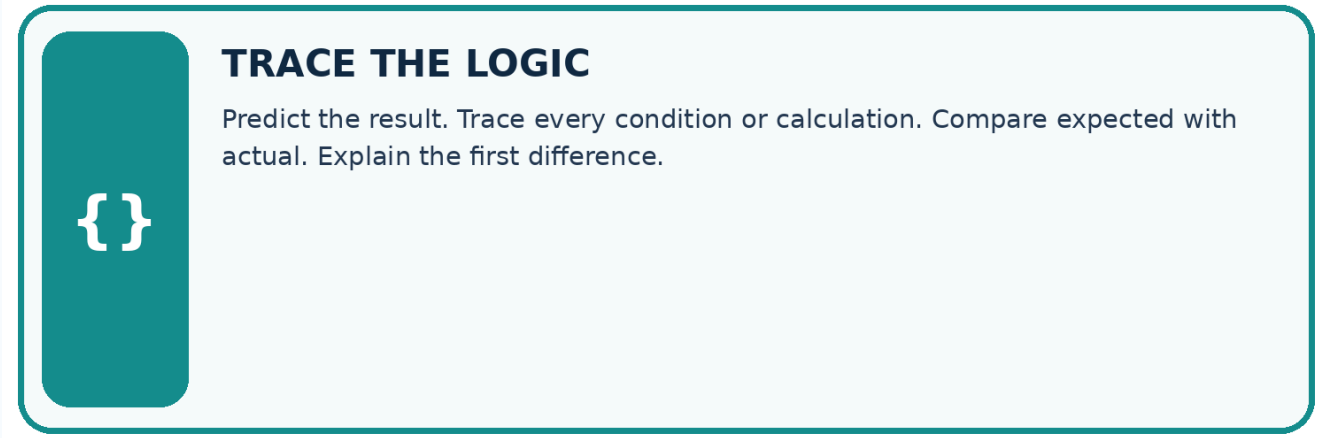
Practise together

> Practise Choose Informative Questions with support, then explain the deciding step in your own words.

LOGIC STUDIO

Trace Before You Run

Block code, pseudocode, tables, and diagrams are different ways to expose the same reasoning.



SET **evidence**_record PREDICT expected_result RUN one normal case and one boundary case COMPARE actual_result with expected_result IF different: locate first cause; change one factor; RETEST EXPLAIN result, limitation, and next safe step

Trace one **input**. Which line changes the state or chooses a path?

Write or draw your reasoning and evidence.

A closer explanation

Do not assume near=1 must always predict blocked. If a new **training** set gives that **branch** a clear majority, this same learner will store clear there. A feature name is not the answer.

Use the **training** labels belonging to that **branch**.

Adding more questions can fit finer **training** details but does not guarantee better new-case performance. Record any **depth** choice made during development, and keep final test data separate.

A **leaf** majority describes this **training** group; it is not a promise about every future example.

Practise with less help: Verify the candidate scores

Use workbook W2 for the connected example “**Fitting** a decision tree”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for **Fitting** a decision tree. Proposed explanation: “A near **branch** must predict blocked”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Choose the best original question and explain why its more balanced **split** matters here.

> **My explanation using a pictured step and the given facts**

This is supported practice. Keep the separately named independent case for an attempt before feedback.

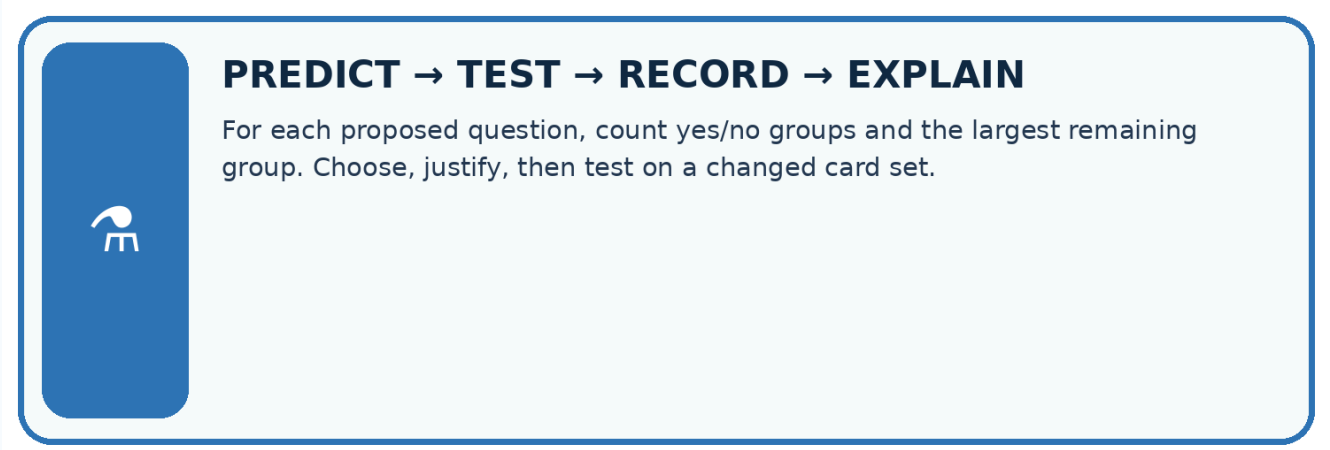
Run the investigation

> Investigate Choose Informative Questions. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



For each proposed question, count yes/no groups and the largest remaining group. Choose, justify, then test on a changed card set.

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

> **Test / input:** normal
> **Prediction:** _____
> **Actual:** _____
> **Evidence / calculation:** _____
> **Change or keep?:** _____

> **Test / input:** boundary

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: missing

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: fresh transfer

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

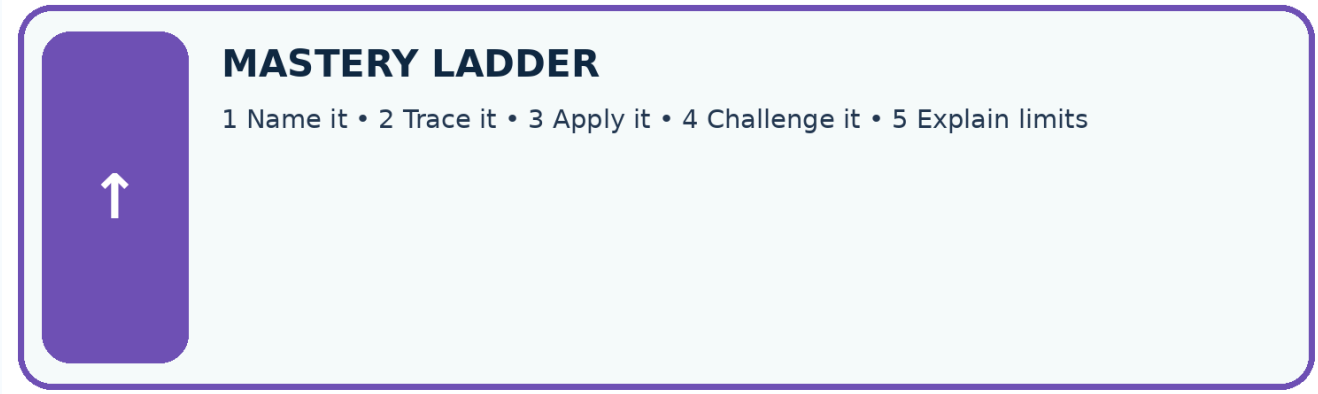
Build your understanding

> Move from recall to tracing, diagnosis and transfer in Choose Informative Questions. Explain why each step is justified.

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the worked example.



1 • NAME: Define the concept without repeating the textbook.

2 • TRACE: Show the information, state, **branch**, calculation, or **evidence** path.

3 • APPLY: Solve a different example independently.

4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.

5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

My strongest **evidence** of independent understanding:

> **Write or draw your reasoning and evidence.**

STUDENT LESSON · 11 OF 18

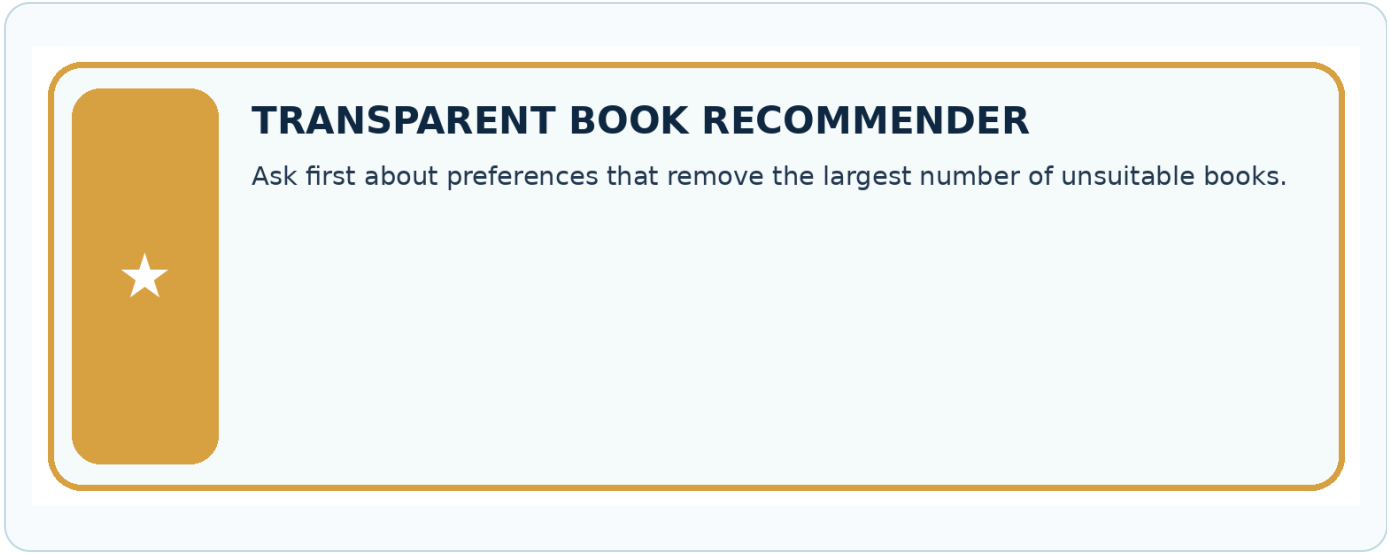
Connect your project

> Apply this week's idea to Transparent Book Recommender. Keep an annotated decision and the evidence supporting it.

CUMULATIVE PROJECT

Transparent Book Recommender

Every week adds one inspectable piece. The system must show how it reached a result.



THIS WEEK'S CHECKPOINT Ask first about preferences that remove the largest number of unsuitable books.

Trace the system

SYSTEM MAP Reader need → allowed **input**s → transparent rule or score → ranked candidates → explanation → test record → **human decision**

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under **human control**?

Write or draw your reasoning and evidence.

Responsible design

RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse a ranking with a fact about a person.

Evidence for your project

Choose recommender questions that distinguish feasible options, documenting the decision criterion rather than asking unnecessary personal details.

STUDENT LESSON · 12 OF 18

Show what you know

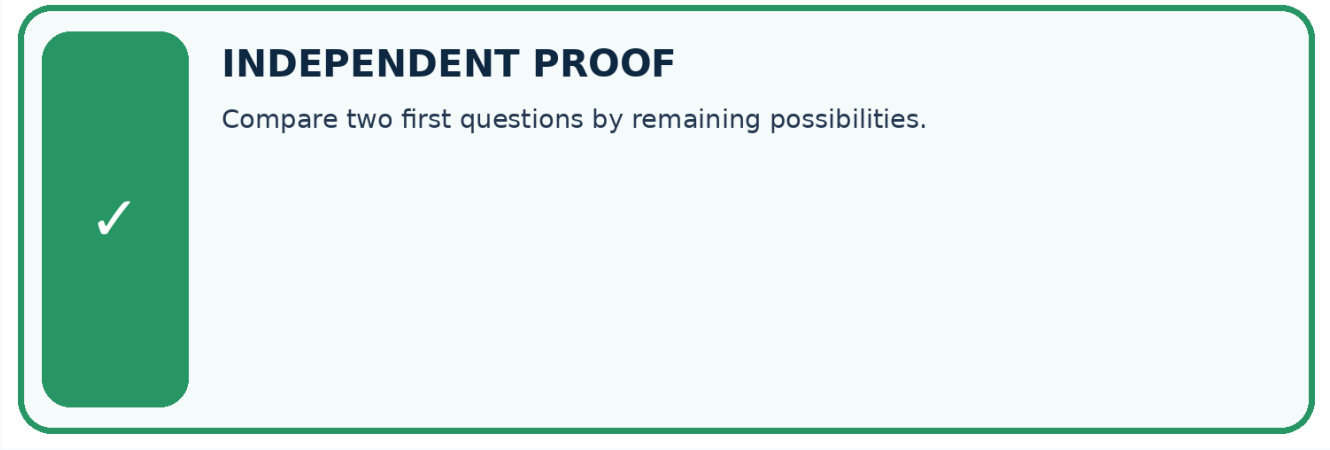
Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

Independent evidence

INDEPENDENT MASTERY PROOF

Explain • Transfer • Test • Defend

Complete this page without copying the worked example. Your reasoning matters more than decoration.



Try a more demanding case

CHALLENGE Compare two first questions by **remaining possibilities**.

1. My fresh example, trace, calculation, or design:

Write or draw your reasoning and evidence.

2. My **evidence** that it works (include a boundary or missing-data case):

Write or draw your reasoning and evidence.

3. One limitation, fairness concern, or situation needing human review:

Write or draw your reasoning and evidence.

Check your explanation

SELF-CHECK ☐ I explained the **mechanism** ☐ I used **evidence** ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

STUDENT LESSON · 13 OF 18

Study a complete case

Revisit the complete Choose Informative Questions case for supported practice. Repeated examples do not become fresh assessments.

Week 15 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

Eight blue cards contain four circles and four triangles; target is shape.

The question

> Which question is more useful: blue or circle?

Worked explanation

Circle separates target groups. Blue puts every card in one **branch**. Usefulness depends on the task and examples.

Explain it in your own words

> Which detail supports the answer, and why does it matter?

> **Write or draw your reasoning and evidence.**

Trace the supplied case visually

Choose a question that separates the target

Evidence or stage	Value or result
Blue?	All eight stay together
Circle?	Four circles versus four triangles
Target	Shape, not colour

Eight blue cards contain four circles and four triangles; target is shape.

> Which question is more useful: blue or circle?

Circle separates target groups. Blue puts every card in one **branch**. Usefulness depends on the task and examples.

STUDENT LESSON · 14 OF 18

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Changed independent case

Ten cards have six pet subjects and four space subjects; all are green. Compare asking “green?” with asking “pet?” to identify a card. Give the largest remaining group and one limitation.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 15 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

Cards are two cats and two dogs, all red. Which question helps separate species: is it red or does it bark? State your assumption.

My answer, with a trace, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Ask first about preferences that remove the largest number of unsuitable books.

The **evidence** I will keep

> **Write or draw your reasoning and evidence.**

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Transfer practice from the original programme

> Six cards contain three animal and three space books; all have blue covers. Compare asking 'blue cover?' with 'animal subject?'. Is the better question guaranteed to work on inaccurately tagged books?

> **Explain the mechanism, evidence and one limit.**

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Choose Informative Questions, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: A fictional guessing game has eight catalogue cards and asks yes/no questions to narrow the target.

By the end, I can	How I will show it
Fit a root question and leaf predictions from training examples	Explain the deciding step, show a trace or test, and state one limit.

By the end, I can	How I will show it
Trace a new example through the learned tree	Explain the deciding step, show a trace or test, and state one limit.
Compare a shallow tree with a baseline and state its limits	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

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Finish and return

> Explain Choose Informative Questions without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Fit a **root** question and **leaf prediction**s from **training** examples

My deciding step, evidence and one remaining question

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional guessing game has eight catalogue cards and asks yes/no questions to narrow the target.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Yes 8; No 0; largest remaining group 8.

1. List possibilities

2. Mystery question

3. Paperback question

4. English question

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: English question Yes 8; No 0; largest remaining group 8.

Under the stated worst-case criterion, mystery is the most useful of these questions because it leaves at most four cards. The constant language clue never narrows the set.

This is an identification-game criterion; a classifier **fitting** labels may instead compare **prediction** errors or another declared **training** measure.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: The current possibility set contains only the four mystery cards. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. Asking mystery now leaves all four; it no longer narrows the set.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Asking mystery now leaves all four; it no longer narrows the set. Question value depends on the remaining candidates, not just on its wording.

Repair a missing building block

Review Decision Trees (ai-2/14). Goal: Fit a **root** question and **leaf prediction**s from **training** examples. Short practice: Why does changing colour alone not change this tree's **prediction**?.

Return to this week and explain what you changed.

Review Accuracy as a Fraction (ai-2/12). Goal: Explain why overall accuracy can hide failure on a less common class. Short practice: Why is 100% repair detection not sufficient to declare the changed classifier best?.

Return to this week and explain what you changed.

Review Features and Distractors (ai-2/9). Goal: Choose recorded properties that preserve distinctions needed for a stated task. Short practice: Does this one pair prove the classifier always ignores subject?

Explain a narrower claim.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

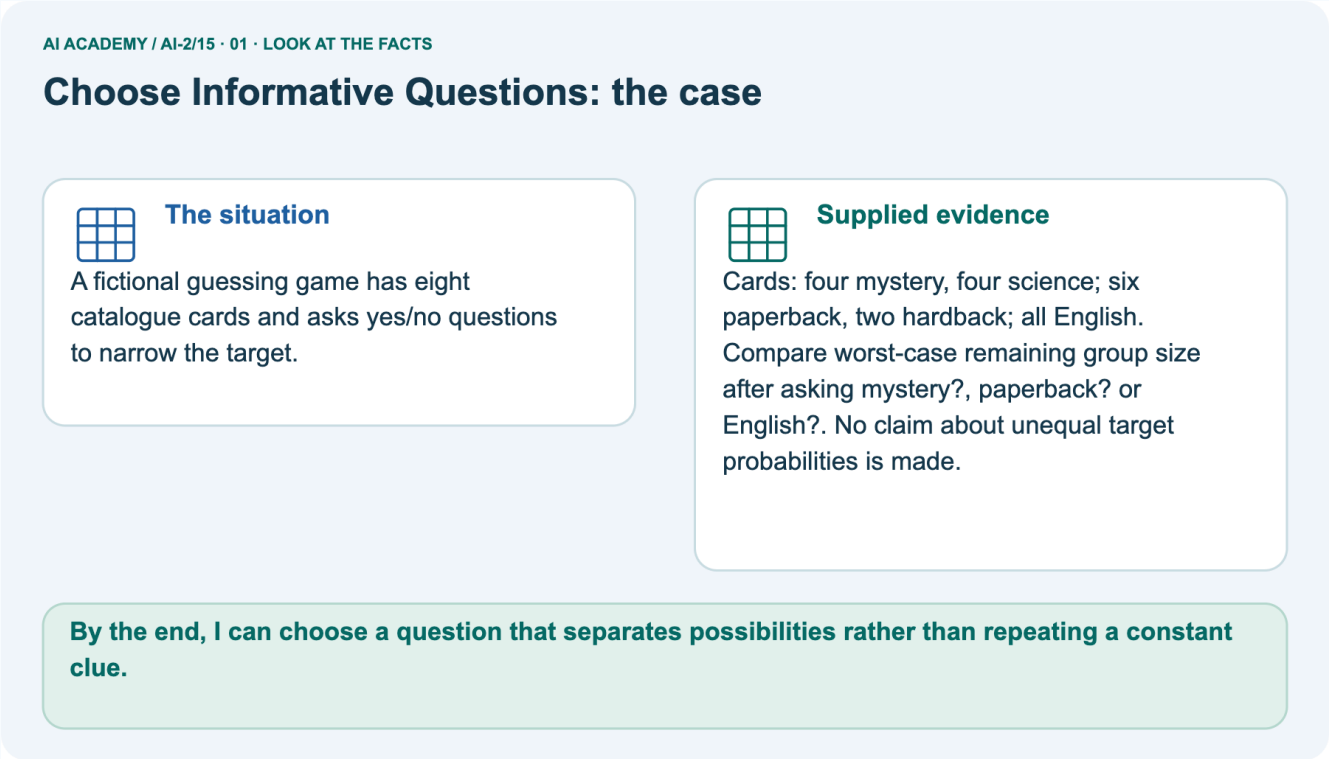
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Choose Informative Questions graphic using its labels, deciding evidence and relevant limitation.

01. Choose Informative Questions: the case



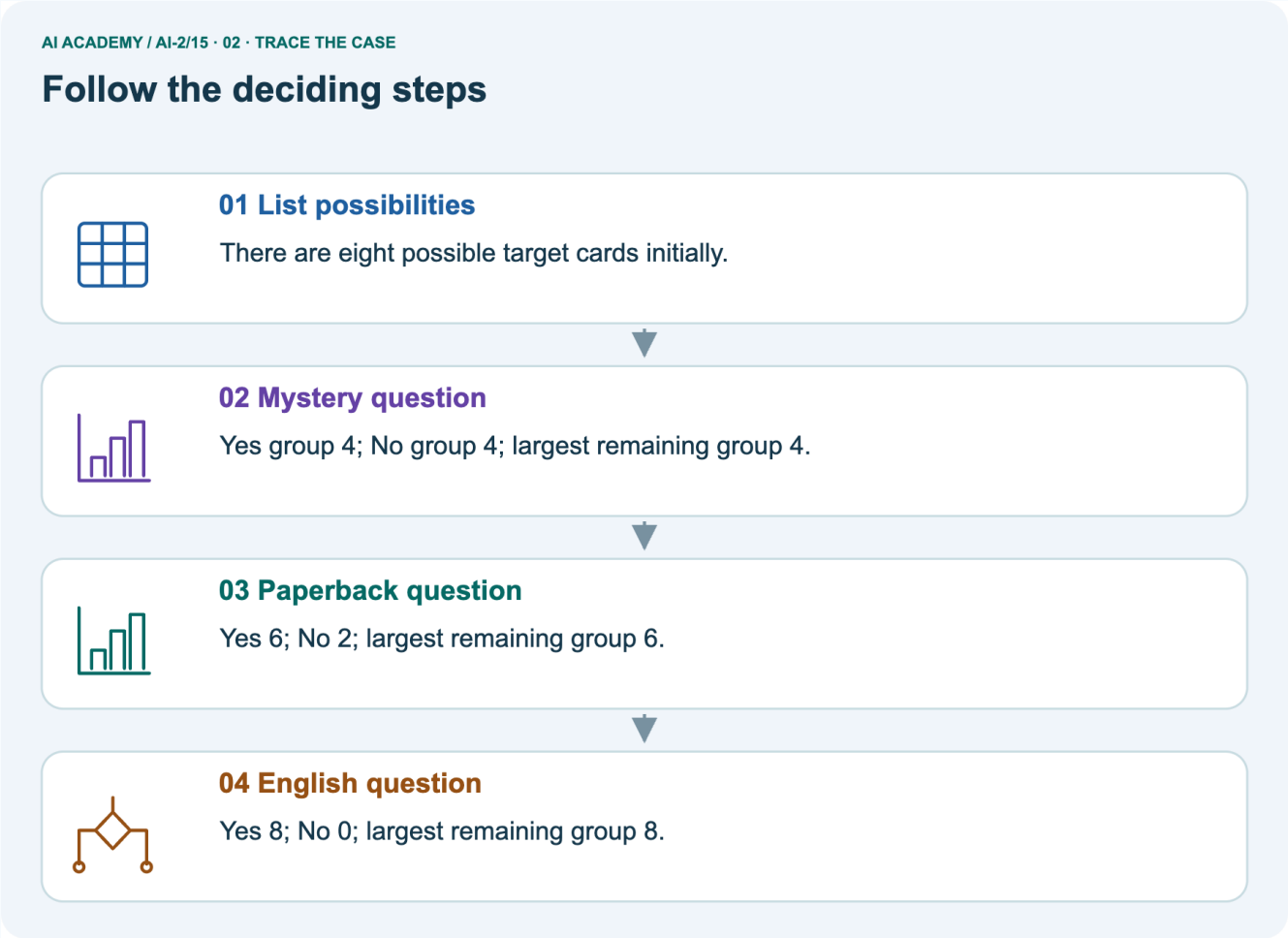
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional guessing game has eight catalogue cards and asks yes/no questions to narrow the target. Supplied facts: Cards: four mystery, four science; six paperback, two hardback; all English.

Compare worst-case remaining group size after asking mystery?, paperback? or English?. No claim about unequal target probabilities is made.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

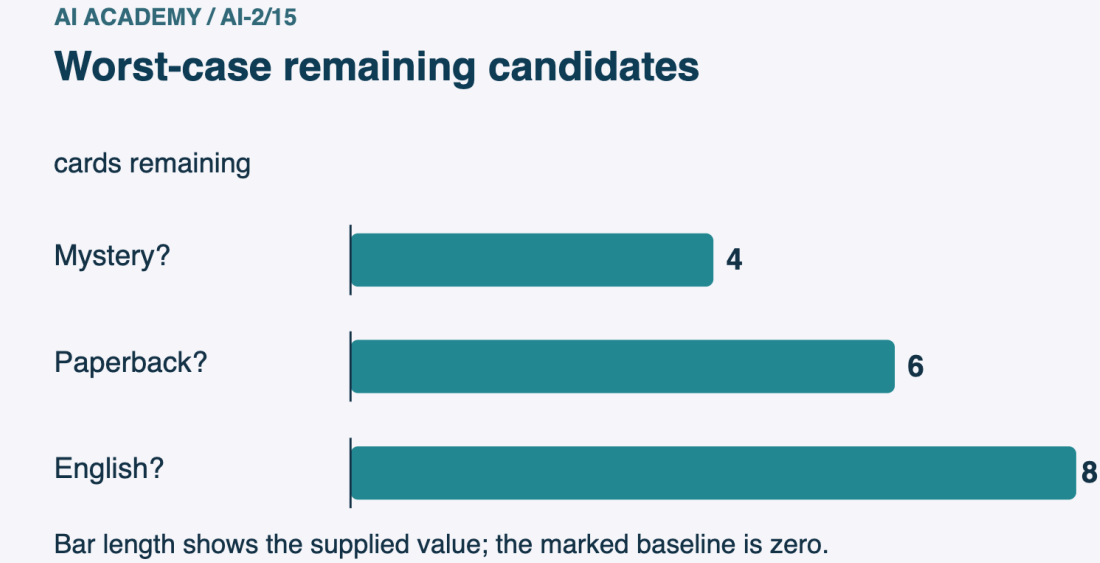


Follow the labelled picture with the matching explanation. Open the vector diagram

List possibilities: There are eight possible target cards initially. → Mystery question: Yes group 4; No group 4; largest remaining group 4. → Paperback question: Yes 6; No 2; largest remaining group 6. → English question: Yes 8; No 0; largest remaining group 8.

> Point to each numbered stage and explain which recorded **input** it uses.

03. Worst-case remaining candidates

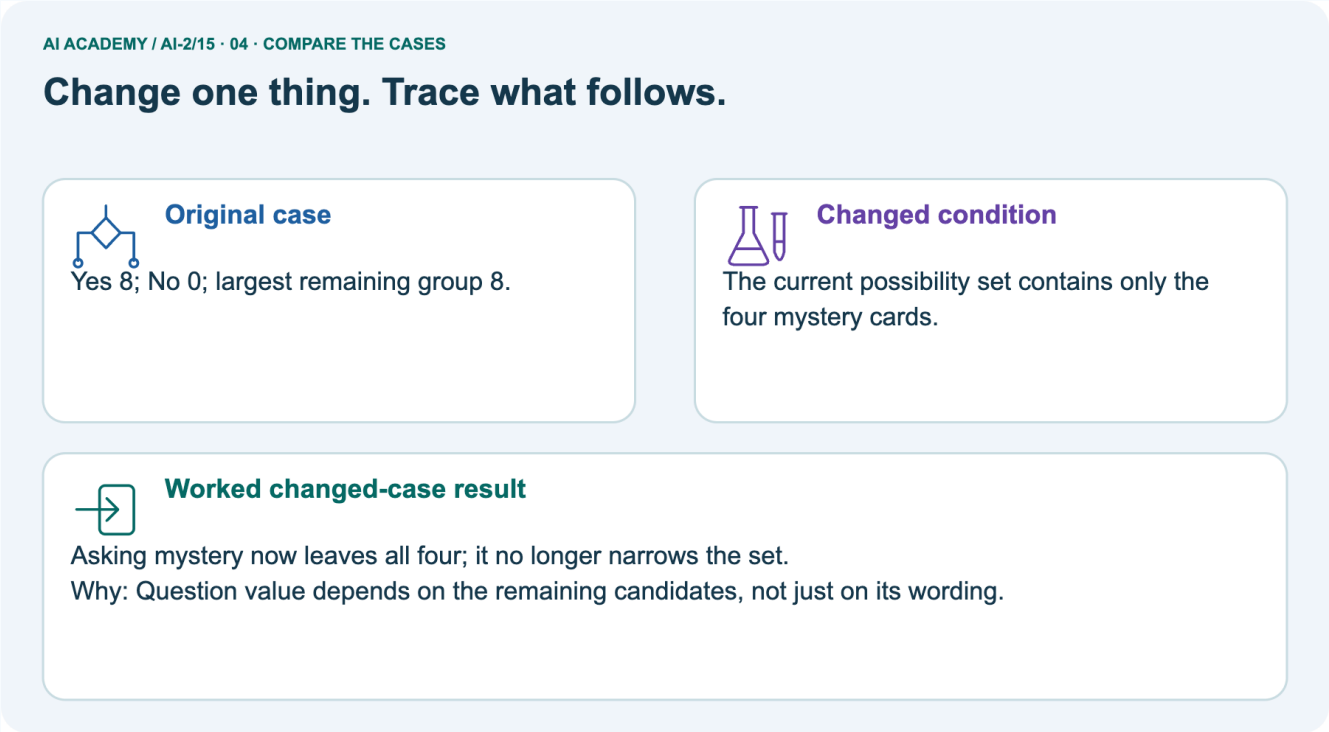


Follow the labelled picture with the matching explanation. Open the vector diagram

Worst-case remaining candidates. Units: cards remaining. Mystery?: 4. Paperback?: 6. English?: 8.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



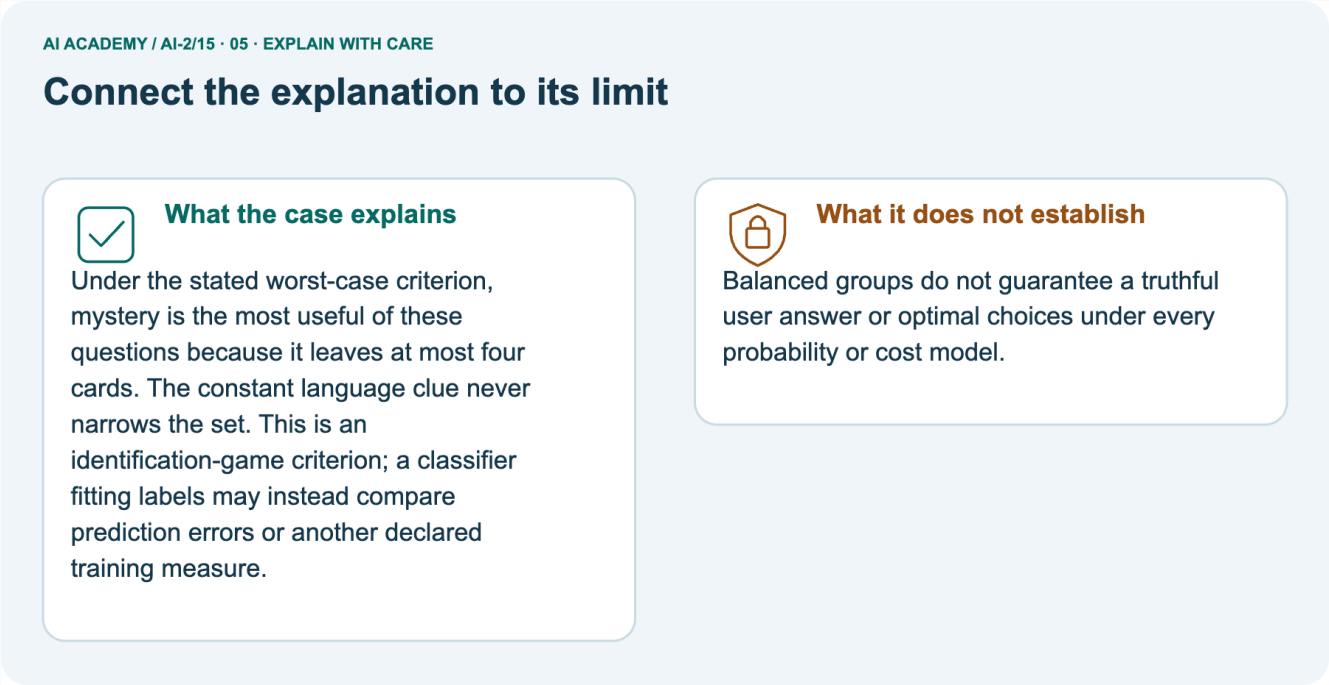
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Yes 8; No 0; largest remaining group 8. Changed condition: The current possibility set contains only the four mystery cards. Expected result: Asking mystery now leaves all four; it no longer narrows the set.

Reason: Question value depends on the remaining candidates, not just on its wording.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

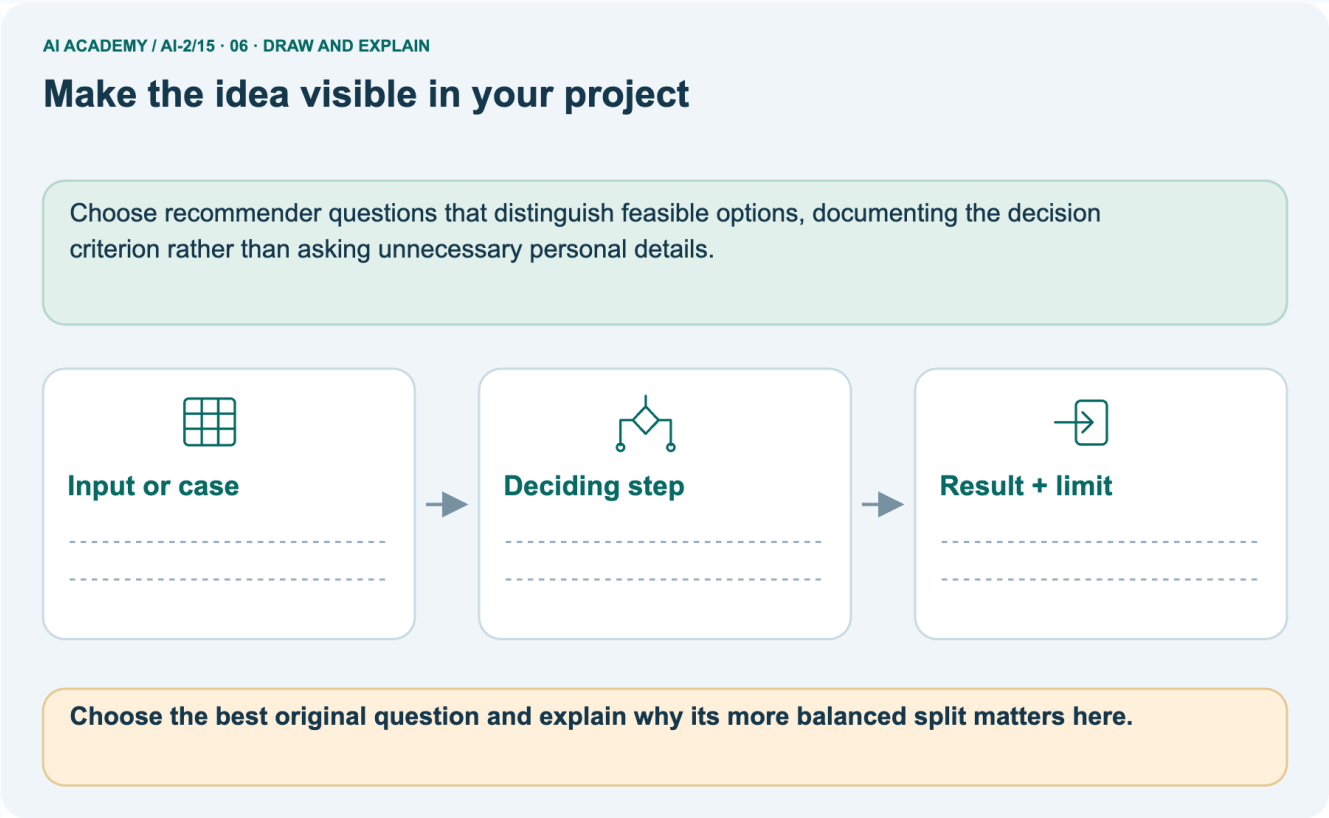
Under the stated worst-case criterion, mystery is the most useful of these questions because it leaves at most four cards. The constant language clue never narrows the set.

This is an identification-game criterion; a classifier **fitting** labels may instead compare **prediction** errors or another declared **training** measure.

Limit: Balanced groups do not guarantee a truthful user answer or optimal choices under every probability or cost model.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Choose recommender questions that distinguish feasible options, documenting the decision criterion rather than asking unnecessary personal details. Draw your **input**, deciding step, result and limit.

Supported question: Choose the best original question and explain why its more balanced **split** matters here.

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.